COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 9 : SLO – 1 | AND, OR, NOT gates
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II. Multiple Choice Questions

• 1. b — All inputs are 1
• 2. c — All inputs are 0
• 3. c — Y = A■
• 4. b — ■
• 5. b — 1

I. Match the Following

• 1. AND gate output when all inputs are 1 — 1 (HIGH), from Y = A■B (b)
• 2. OR gate Boolean expression — a) Y = A + B
• 3. NOT gate symbol — c) ¯ (bar)
• 4. AND gate symbol — d) ■
• 5. OR gate output when all inputs are 0 — e) 0

III. Activity Question

• AND gate: Y = A■B — output 1 only when ALL inputs are 1. Symbol: D-shape with flat input side. Truth table: 00→0, 01→0, 10→0, 11→1. Example: two series switches lighting one lamp.
• OR gate: Y = A+B — output 1 when ANY input is 1. Symbol: curved input side, pointed output. Truth table: 00→0, 01→1, 10→1, 11→1. Example: doorbell wired to two parallel push-buttons.
• NOT gate: Y = A■ — output is the inverse of the input. Symbol: triangle with inversion bubble. Truth table: 0→1, 1→0. Example: light that turns ON when a dark sensor reads 0.